



Mushroom-Identifying App

There is a new mushroom identifying app and experts are calling it 'potentially deadly'.
<http://dgit.in/MushroomAp>



Assistant for ChromeOS

Reports from Chrome Unboxed suggests that Google is potentially removing 'OK, Google' from ChromeOS. <http://dgit.in/OKchrome1>



GIGABYTE X299 AORUS Gaming 9



ASUS PRIME X299-DELUXE

FEATURES

GIGABYTE X299 AORUS GAMING 9

We see five physical PCIe x16 slots allowing for multi-GPU configurations involving double and triple slot graphics cards. There are 6 USB 3.1 ports with one of them being Type-C and 4 USB 3.0 ports on the rear I/O. However, we see no USB 2.0 ports on the rear but there are headers on the motherboard should you need them. We also see 8 SATA III connectors in total and 3 M.2 connectors but no U.2 anywhere. However, the packaging does include a U.2 adapter. The onboard Wi-Fi is a dual band 802.11ac with peak data rate of 867 Mbps.

ASUS PRIME X299-DELUXE

There are four PCIe x16 physical slots on this board with 3-way multi-GPU support. However, their placement dictates what kind (width) of graphics cards you can use for your multi-GPU configuration, thus, reducing flexibility. But we see 2 PCIe x1 slots thrown in, so there's more value addition. There are 4 USB 2.0 ports on the rear I/O and 2 more via headers. As for USB 3.1 there are 5 ports in total, one being Type-C. We see 7 SATA III ports and the last one repurposed for U.2. Also, this board has 802.11ad with a peak rate of 4,600 Mbps.

PERFORMANCE

GIGABYTE X299 AORUS GAMING 9

In AIDA64's memory bandwidth test, we saw copy speeds being almost the same. The difference arose in write and read speeds. The Gaming 9 has a slightly better write speed compared to the PRIME X299-DELUXE, giving rise to a difference of 2.8%. We then moved on to the audio chipset which is a Realtek ALC1220 along with an OPA1622 op-amp and an ES9018K2M DAC. Running Rightmark totally tilted the scales in favour of the Gaming 9 as we see reduced noise, reduced cross talk and much better frequency response across the entire spectrum.

ASUS PRIME X299-DELUXE

While it didn't lead in write speeds, read speeds were slightly greater on this board by 2%. Every other software performance benchmark that we ran was inconsequential in terms of difference between the two boards. In audio, we see the same Realtek ALC1220 audio chip but an amp or a DAC are absent. Given the simple implementation of the audio circuitry, we noticed that the audio performance was lower comparatively. THD and IMD were the only aspects where the two boards were close. It goes without saying that the PRIME X299-DELUXE was better with regards to Wi-Fi performance.

CONCLUSION

We have two really well-built motherboards from GIGABYTE and ASUS with us. Both having a really good VRM design albeit, the ASUS one does draw CPU power from an 8+4 pin power connector while the GIGABYTE board leaves more headroom with the dual 8-pin connectors. The Gaming 9 and PRIME X299-DELUXE have ample amount of accessories to make up for features that couldn't have been included on the board directly. Though, we feel that by not placing the less popular U.2 on the board and instead giving an extra SATA port was a better decision. Also, the audio solution on the GIGABYTE board is much better. Let's not even get into the RGB aspect. Moving on, we have the Wi-Fi which obviously works in favour of the ASUS board. They've also included a Thunderbolt 3 card and a

GPIO cable for the same with a few temperature probes as well. The AORUS Gaming 9 has a Thunderbolt AIC connector but instead of giving an AIC for TB3, they've given a PCIe adapter for M.2 cards with a heavy duty heatsink. It stands to reason that the ASUS board offers a wider feature set but the GIGABYTE board focuses on higher performance from its existing feature set. This simply goes on to say that there's more value for money with the ASUS motherboard but if you want better performing audio solution and more high speed USB ports then the GIGABYTE board should be your choice. The ASUS PRIME X299-DELUXE costs ₹39k while the GIGABYTE board costs upwards of ₹50k. This reiterates our earlier statement of getting more for your money with the ASUS board. 